

Real-World Evidence / HEOR

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 07 of 8 · build cohorts and translate code so analysts analyze — an epidemiologist approves the study

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM WRANGLING TO EVIDENCE

Real-World Evidence / HEOR

A governed RWE/HEOR copilot that defines cohorts, translates between code systems, and assembles auditable data lineage — so analysts spend time on analysis, while an epidemiologist or biostatistician approves the study and its conclusions.

~45%

of analyst time goes to data prep, not analysis (measured)

[sector-press/estimate — Anaconda]

~6 months

to manually build a retrospective RWE database study

[sector-press/estimate]

~\$1.3M / yr

non-analytic labor for a 20-person RWE team (modeled)

[modeled]

The Issue & The Cost of Doing Nothing

THE ISSUE

- RWE/HEOR requires cohort definition, confounding control, and structured code translation.
- Analyst time is disproportionately spent on data wrangling rather than evidence generation.
- FDA's RWE Framework hinges on data fitness-for-purpose and auditable lineage — hard to do by hand.
- Follow-on FDA RWD/RWE guidances (2023–24) keep raising the bar on traceability.

THE COST OF DOING NOTHING

~\$1.3M / yr / team

Modeled: analyst ~\$150K fully loaded × ~45% on wrangling = ~\$67K/yr each; a 20-person RWE team ≈ ~\$1.3M/yr of non-analytic labor. A single study can take ~6 months to build before analysis.

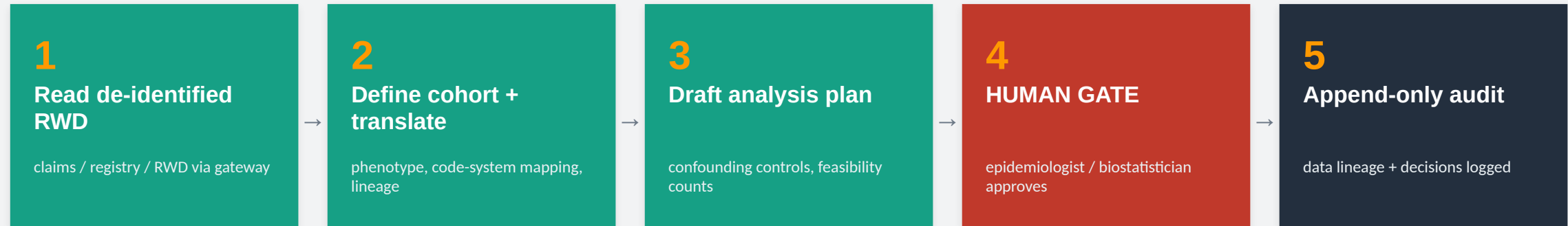
- Slow evidence generation that delays HEOR dossiers and market-access decisions.
- Lineage gaps that weaken the fitness-for-purpose case to FDA.

[modeled — loaded analyst cost × Anaconda data-prep share]

The design line: the agent builds cohorts, translates code, and documents lineage; an epidemiologist or biostatistician approves the study design and conclusions.

How We Solve It — A Governed Pipeline

Read de-identified RWD through a scoped gateway, define cohorts and translate code with full lineage — then route the study to a qualified human.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

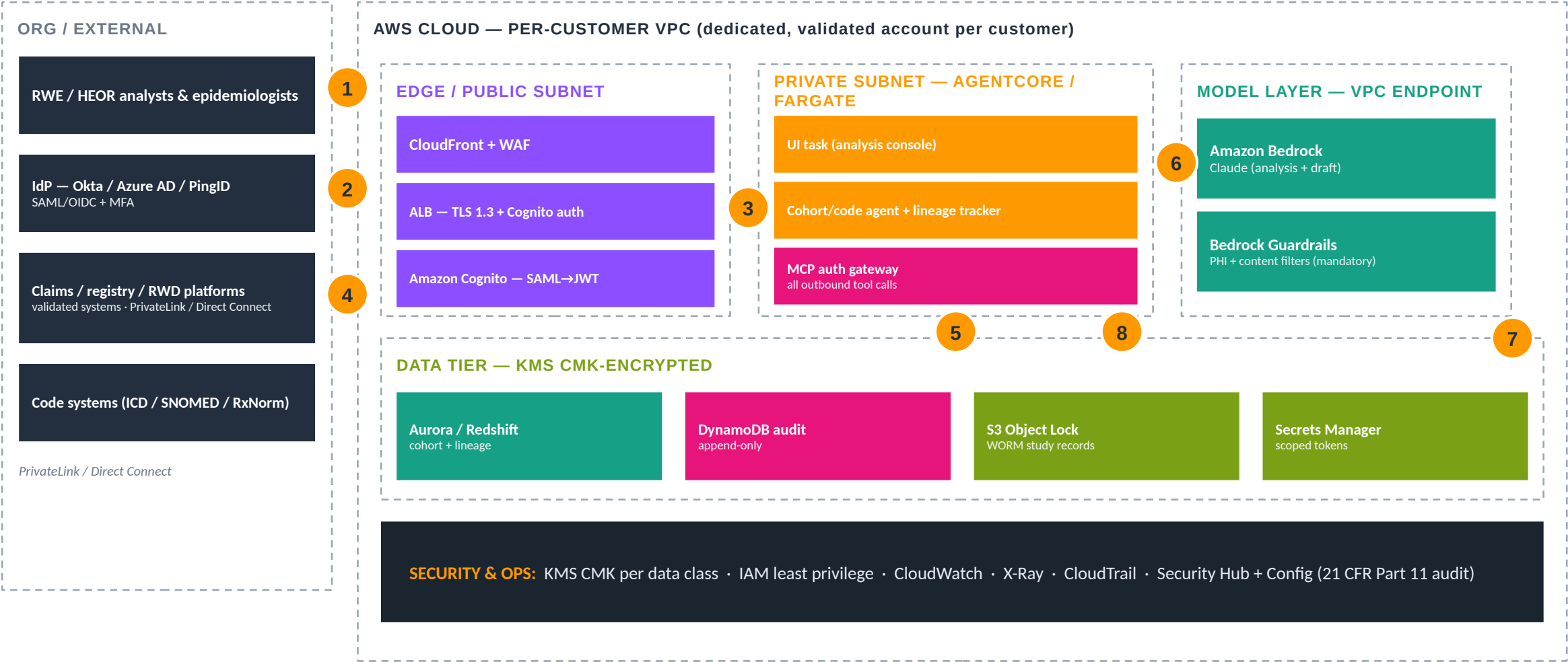
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides the analysis & conclusions

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 analyst sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to analysis console 4 de-identified RWD reads over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails 7 full data lineage + decisions persisted to append-only audit 8 connectors reachable only through the governed MCP gateway (de-identification enforced)

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~45% → 90%

analyst time shifted from prep toward analysis

[design target vs. Anaconda baseline]

code xlat

ICD/SNOMED/RxNorm mapping with provenance

[design property]

lineage

auditable data provenance for FDA fitness-for-purpose

[gov/peer-reviewed — FDA RWE]

hours → min

vendor proxy: AstraZeneca analyst acceleration

[vendor — AWS]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (analyst / epidemiologist roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect de-identified claims/registry/RWD; enforce de-identification at the gateway.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; point connectors at de-identified RWD; wire code-system references.

aws-native-reference/07-rwe-heor/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.